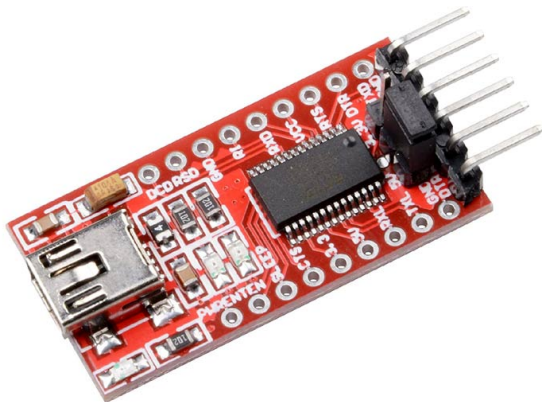


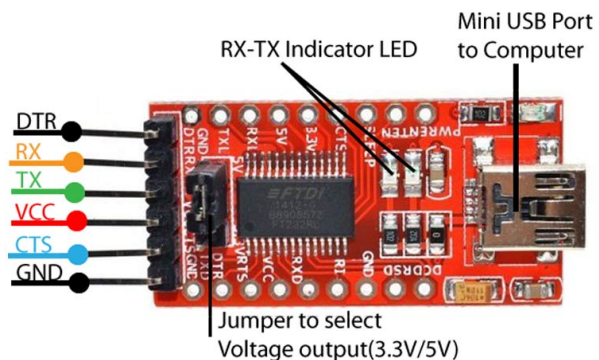
FT232RL USB TO TTL 3.3V/5V FTDI Serial Adapter Module

28 May 2021 - 0 Comments



(/sites/default/files/components/FT232RL-USB-to-TTL-Converter.jpg)

FT232RL USB TO TTL Converter



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FT232RL USB TO TTL Converter Pinout

Description of FT232RL USB TO TTL Converter

The **FT232RL** USB to TTL 3.3V/5V FTDI Serial Adapter Module is a popular module that is used to connect a TTL serial communicating device to a PC via a USB mini port. This converter module has options for different output voltage options which are set by the jumper on the board. FT232RL USB to TTL Serial Adapter Module can be used to upgrade any **legacy peripherals to USB**.

Features and Specifications of FT232RL USB TO TTL Converter

This section mentions some of the features and specifications of the FT232RL USB to TTL Converter:

1. Operating Voltage: 5V/3.3V DC
2. Max Current Draw: 5V - 500mA; 3.3V - 50mA
3. Connector: Mini USB
4. Fully integrated 1024 bit EEPROM storing device descriptors and CBUS I/O configuration
5. Data transfer rates from 300 baud to 3 Mbaud (RS422, RS485, RS232) at TTL levels

6. 128 byte receive buffer and 256 bytes transmit buffer
7. Transmit and receive LED drive signals
8. Fully integrated clock generation with no external crystal required

Alternatives for FT232RL USB TO TTL Converter

FT234XD, UMFT201XB-01 XBee, TTL-232R-3V3, VNC1 Module VDIP1

Note: More technical information can be found in the **FT232RL Datasheet** linked at the bottom of this page.



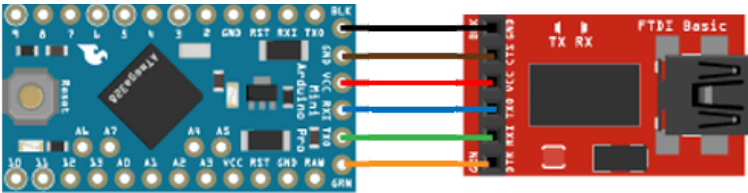
Pin Configuration of FT232RL USB TO TTL Converter

The FT232RL USB to TTL Converter has 6 pins. The table below depicts all the pin types along with the function of each pin.

Pin Type	Description
DTR	Data Terminal Ready(Output used for flow control)
RX	Serial Data Recieve
TX	Serial Data Transmit
VCC	Power Input
CTS	Clear to send(Input used for flow control)
GND	Ground

Programming an Arduino Pro Mini using FT232RL USB TO TTL Converter

A widely used application of an FTDI module of such sort is used to program small microcontroller boards such as Arduino Pro mini as shown in the image. The connections of the FT232RL USB TO TTL Converter to a Pro mini are pretty simple and can be followed as the image shown below.



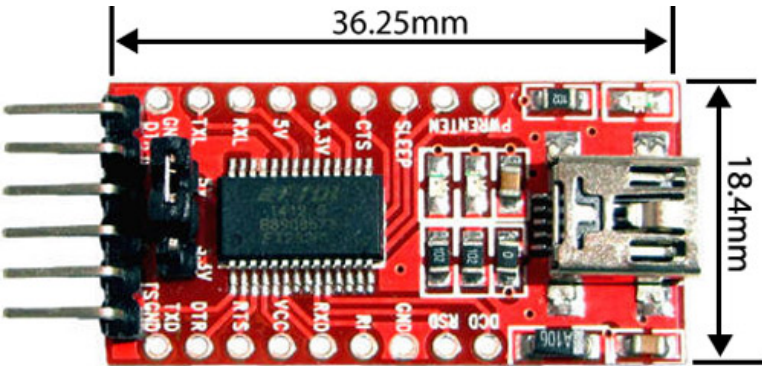
Applications of FT232RL USB TO TTL Converter

Below are the applications of FT232RL USB TO TTL Converter:

- 1. Programming AVR, ARM-based microcontrollers/microprocessors.
- 2. USB to RS232/RS422/RS485 Converters
- 3. USB Smart Card Readers
- 4. For serial communication with GPS devices
- 5. Serial terminal on devices like Raspberry Pi

2D Model of FT232RL USB TO TTL Converter

The 2D model of the FT232RL USB TO TTL Converter can be seen in the image below; the dimensions of the board shown are in millimeters and it can be followed to design custom footprints of the module.



Component Datasheet

FT232RL USB TO TTL Converter Datasheet
(https://components101.com/sites/default/files/component_datasheet/FT232RL-USB-TO-TTL-Converter-Datasheet.pdf)

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